Journal of Cleaner Production 279 (2021) 123340

Contents lists available at ScienceDirect

Journal of Cleaner Production

journal homepage: www.elsevier.com/locate/jclepro

Coastal container multimodal transportation system shipping network design—toll policy joint optimization model

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ARTICLE INFO

Article history:
Received 13 January 2020
Received in revised form
26 June 2020
Accepted 17 July 2020
Available online 11 August 2020

Handling editor: Prof. Jiri Jaromir Klemeš

Keywords:
Sustainable policy
Coastal transport system
Carbon emission
Shipping network design
Toll policy optimization model

ABSTRACT

A coastal container multimodal transportation system has two modes of transportation, i.e., land and water. Due to economies of scale, the latter has become an emerging way to decrease the pollution of the system. This paper establishes a programming model to improve the system's sustainable transportation policy. The model aims to minimize the carbon dioxide emissions of the system by optimizing two types of policies (i.e., a shipping network design policy and a toll policy) simultaneously. A local optimization algorithm is proposed to solve the abovementioned model. Real-world data on container transportation in the Bohai Rim in Northeast China are used to perform two sensitivity analyses. The running time of each calculation experiment does not exceed 2 minutes, showing the efficiency of the aforementioned algorithm. The experimental results show that the policies can reduce carbon dioxide emissions from container transportation in Bohai Rim from 3.29% to 6.70% under different scenarios. The diminishing marginal utility of the government's sustainable policy is also revealed.

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1. Introduction

In the last few decades, the growing awareness of environmental issues has led to an increased effort toward environmental protection (Zhu and Hu, 2019). In the transportation field, the concept of *sustainable transport* has been proposed by various scholars. According to Chiou et al. (2013) and de Campos et al. (2019), the development of low-pollution transport is an effective means of making current transport systems more sustainable. Recently, the governments of various countries have actively promoted the implementation of sustainable policies in coastal container multimodal transportation systems (CCMTSs) around the world. A CCMTS is a complex multimodal transportation system composed of a land transport subsystem and a water transport subsystem (see Chen et al. (2014) for details). In a CCMTS, cargo transportation can be performed on both land and water. The former has the advantage of fast transport and is not susceptible to weather factors, while the latter has the advantage of large shipping volumes and low carbon dioxide (CO_2) emissions per unit cargo per unit distance. Because of their complementary advantages, governments can realize the sustainable transport of CCMTSs through a *transportation mode-shift policy*. In practice, the government usually influences the decision-making behavior of various stakeholders in the CCMTS through policy regulations, thereby changing the pattern of cargo flow distribution in the CCMTS and reducing carbon emissions (see the conclusion of Matajič et al. (2015)).

This paper aims to model the government's optimal sustainable policy implementation plan in CCMTS, which focuses on two common policy types (i.e., network design policy and toll policy). Helping water carriers to redesign the shipping network is a common option for the government. However, the government needs to consider which routes should be added to the operational capacity and then provide a modest investment. For example, the *motorways of the sea program* introduced by the European Union and the *marine highway program* designed by the United States (see Douet and Cappuccilli (2011) and Chen et al. (2014), respectively). In addition, toll policies are the most common policy used in the transportation field. In recent years, the carbon emission toll (also known as 'carbon tax') imposed on transportation systems has become a new trend to reduce emissions (see Oreskes (2011) and

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https://doi.org/10.1016/j.jclepro.2020.123340
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Jian et al. (2017)) partly because 'there is a clear relationship between carbon emissions and the use of fossil fuels, with a direct influence on the various emission sources' (Yang, 2000).

The implementation of the above policies involves three stakeholders, namely, the government, shippers and land/water carriers. Different policy implementation plans will have a direct impact on the behavior of all stakeholders. One can see that this decision problem involves both macrolevel issues such as system design and microlevel issues such as individual behavior choices. These different levels of factors are related to and influence each other, which constitutes a complex decision-making problem, which is called the shipping network design—toll policy joint optimization problem of CCMTSs (short as STNP).

We make three main contributions in this paper. (i) A programming model is established that takes the lowest CO_2 emissions of the CCMTS as the objective and fully considers the conflicts among three stakeholders. (ii) The government's network design policy and toll policy can be optimized simultaneously. (iii) Real world data are used to explore the government's optimal policy scheme.

The rest of this paper is organized as follows. We review the relevant literature in the next section. Section 3 gives a description of STNP and explains the assumptions, notations, mathematical description of the transportation network, and link cost functions. Section 4 establishes a model for solving the STNP. Section 5 designs an algorithm for solving the model based on the framework of the active set algorithm (ASA). Section 6 conducts numerical experiments to verify the validity of the model and algorithm. Finally, Section 7 concludes this paper.

2. Literature review

At present, many scholars have focused on sustainability analysis and system efficiency and utility assessment (see Chowdhury et al. (2019a), Chowdhury et al. (2019b) and Chowdhury et al. (2020) for three examples). This paper discusses the sustainability of the transportation system from the perspective of government policy design. The STNP is a composite problem that considers both the shipping network design problem (NDP) and the toll problem (TP) in the multimodal transportation network. In this section, we provide an overview of the literature on these two types of problems as well as the literature that considers policy-network design joint optimization in multimodal transport networks.

The first category is the literature related to the NDP. Backmann first proposed the NDP in 1955 (Beckmann et al., 1955), and after 65 years of continuous development, the relevant literature has become very rich. Systematic reviews of different periods for the NDP can be found in Magnanti and Wong (1984), Bell and Iida (1997), Yang and H. Bell (1998) and Guihaire and Hao (2008). According to these studies, the NDP can be broadly divided into two categories: the discrete network design problem (DNDP; see Wang and Hong (2010)) and the continuous network design problem (CNDP; see Abdulaal and LeBlanc (1979)). Since the shipping network (route) design problem is discrete, the STNP should belong to a special class of DNDP. In recent years, many scholars have considered the factors of sustainable development in DNDP. For example, Wang and Hong (2010) developed a multi-objective mixed integer-programming model to minimize greenhouse gas emissions, fuel consumption and noise in the design of airline hub-and-spoke networks. Lin et al. (2017) proposed a bilevel programming model for railway network design. This model optimized the construction of the railway network under the consideration of investment, transportation costs and environmental costs. Cariou et al. (2018) established a mixed-integer linear programming (MILP) model to maximize the profit by selecting the ports to be visited, the sequence of port visits, the cargo flows between ports, and the number/operating speeds of vessels on each arc of the selected route. According to the above literature, both the DNDP and the CNDP are mainly modeled by a MILP model or a bilevel programming model, with the solution often passing through a heuristic algorithm (such as a simulated annealing algorithm or a GA). Although the use of heuristic/evolutionary algorithms has overcome the difficulties of directly solving the model, the robustness of such algorithms is not strong, and the speed of the solution is closely related to the size of the problem.

The second category is the literature related to TPs. In general, toll schemes can be divided into two main categories (i.e., first-best and second-best schemes, see the classification of Lawphongpanich and Yin (2007)). The first-best scheme assumes that all links in a transportation network need to have a toll scheme (see Dial (1999) and Bai et al. (2004) and Yildirim and Hearn (2005)), while the second-best scheme assumes only that some of the links in the transportation network can be charged (such as for political reasons, e.g., Bard (1998) and Verhoef (2002)). Since we assume that only some of the links can be charged, the STNP belongs to a special class of second-best-scheme problems. Similar to this paper, the literature in recent years considering both toll and environmental factors has also been very rich. For example, Chiou et al. (2013) proposed two bilevel programming models (i.e., an operational model and a planning model) for intercity passenger transport systems under sustainability contexts and solved them by a GA. Wen and Eglese (2016) developed a bilevel pricing model to minimize the equivalents of CO_2 emissions and the total travel time and tested the validity of the model through a small-scale example. Jian et al. (2018) optimized a freeway's truck toll by weight policy considering the external environmental cost. They conducted research on the emissions of Heilongjiang Province in China through field investigations and obtained a policy implementation plan. The abovementioned literature has often aimed to improve traffic conditions (e.g., congestion in transportation systems) or environmental conditions (e.g., CO_2 emissions) in a specific region and to establish different optimization models to determine the optimal toll policy for all (or some) links in the study area. These models generally start from the UE theory and consider the interaction of the choice behavior between different shippers but often do not consider the interaction between different types of carriers in the transportation network. In the STNP, both land and water transport can facilitate the transportation of cargos. Only through the latter can the sustainability of the transport system be improved. Therefore, in addition to the noncooperative game among shippers, different types of carriers in the STNP have a noncooperative game that must be considered.

The third category is the literature related to policy-network joint design in multimodal transport networks. Among different NDPs, multimodal transportation network problems are a series of research hotspots. It gradually developed from the hub-and-spoke problems first proposed by O'kelly (1987). A systematic literature review can be found in Crainic et al. (2018). In recent years, some scholars have also considered research on government tax/subsidy policies and the design of multimodal transport networks. For example, Zhang et al. (2015) discussed an optimization model for transportation network design and the carbon tax rate. The authors extended a bilevel network optimization model and used a GA to solve the model. In their model, the competition between different modes of transport is not considered, and the speed of solving the problem is relatively limited. Recently, Jiang et al. (2020) established a logistics network design model considering CO_2 emissions and the subsidy (which can be understood as a 'negative charge'). The authors discussed the construction plan of the distribution centers and proposed an improved adjustable robust optimization

method to address demand uncertainty. Unfortunately, the authors did not consider the impact of different shippers on the choice of transportation route, nor did they consider the expansion of link capacity.

To the best of our knowledge, no previous study has completely mapped the STNP we proposed. The research gap lies in how to establish a shipping network design—toll policy joint optimization model considering the decision-making behavior of various stakeholders. Therefore, we establish a model based on the framework of mathematical programming with equilibrium constraints (MPEC) to explore the STNP further.

3. Problem description

3.1. STNP

As mentioned above, STNP involves a complex decision process consisting of two types of policy implementations. The *core interests* and *decision-making preferences* of various stakeholders in the CCMTS cannot be ignored (Chiou et al., 2013). The former refers to the issues of greatest concern to the stakeholder, while the latter refers to the constraints to consider when making decisions by a stakeholder. The relationships among the three stakeholders in the CCMTS are shown in Fig. 1.

Generally, the core interest of the government is to achieve the sustainable development of the CCMTS. The implementation of the policies must reduce the total carbon emissions of the CCMTS. The government's decision-making preference is to reduce the carbon emissions of the CCMTS by helping the water carriers redesign the shipping network (i.e., improve the operational capacity) and by designing a toll policy implementation scheme. Note that the investment cannot exceed the budget ceiling. The carriers' core interest lies in obtaining a profit. Their decision-making preference is to determine the freight rate to maximize their own profits based on the current cargo flow distribution in the CCMTS. The core interest of shippers is the stability of the freight rate. Although the toll policy allows the government to recover some of the costs, all shippers should only be slightly affected in terms of the freight rate when compared to the situation without any toll policy, i.e., obeying the *approximately Pareto-improving* principle. Their decision-making preference lies in maximizing their own utility, i.e., each shipper will choose the carrier (transportation mode) and transportation route in the CCMTS freely according to their own utility (i.e., generalized transportation cost).

Based on the above discussion, STNP can be described as a government decision-making problem for achieving sustainable transport in a CCMTS. The goal of the STNP is to minimize the CO_2 emissions of the CCMTS. The constraint that needs to be considered is the government's budget ceiling and the commitment to the shippers' approximately Pareto-improving solution. The decision variables are the implementation plan of the policies. The research questions that this paper can systematically solve include the following.

(i) How can we describe the CCMTS cargo flow distribution considering the multiple stakeholders' choice behavior?
(ii) How should the government optimize the transportation network design scheme to ensure the maximum sustainability of regional transportation?
(iii) How should the government simultaneously formulate a toll policy but in which shippers are only slightly affected?

3.2. Basic considerations

To facilitate the description of STNP and modeling convenience, we first list the following basic considerations (or assumptions).

(i) In the CCMTS, we do not discuss the number of water carriers, herein assuming only that all the water carriers are in a cooperative state. Land transport has numerous land carriers, and they compete with each other.
(ii) The government's toll policy will affect the transportation freight rate as an 'additional price'.
(iii) A shipper will choose the transportation mode/path according to the principle of maximizing their own utility (considering both the freight rate and additional price), that is, following the UE principle. When the cargo flow of each link increases, the carrier will increase the cost of the link. That is, the behavior of the shippers will affect each other, and the carriers will adjust the price according to the shippers' choice.
(iv) The government can add more ships based on the original routes. Thus, the government can improve the customer service level of water transport by expanding the transportation capacity of waterway links, thereby attracting more shippers to use water transport. For the convenience of modeling, we assume that each route adds only container ships of the same size as existing ships and that the capacity can be expanded up to four times the original. The cost of

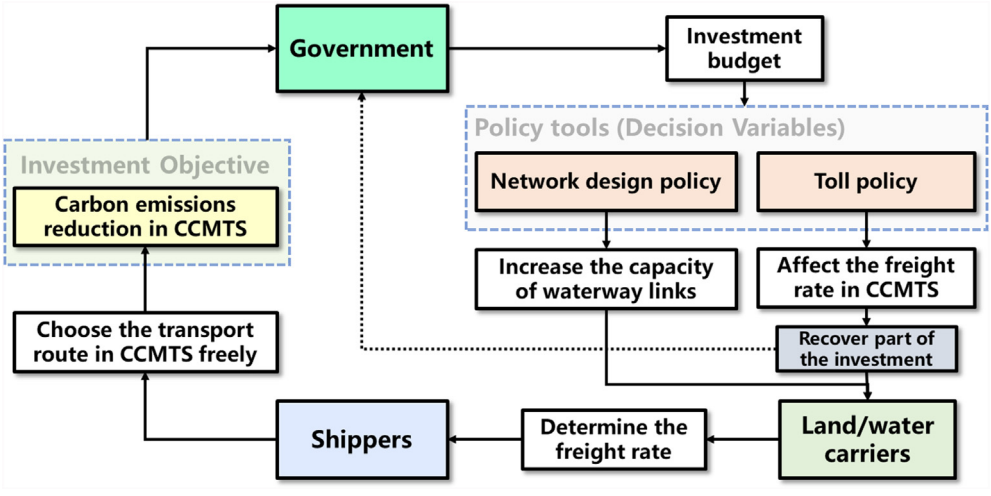


Fig. 1. Game relationship between various stakeholders in coastal transportation networks.

purchasing/leasing ships will be paid by the government. Note that we do not consider the government opening routes that did not exist before, mainly to avoid operational and investment risk.

(v) In the CCMTS, the government can receive funds back through the toll policy. The formulation of the toll policy should not cause a major change in the transportation cost; that is, the solution obeys the principle of *approximate Pareto-improving*.

(vi) The government's goal is to redesign the water subnetwork ('WSN') and implement a toll policy simultaneously with the objective of maximizing CO_2 reductions under the constraints of the established budget ceiling.

3.3. Notations

For the convenience of the reader, the abbreviations frequently used in this paper are listed as follows (in order of appearance).

Abbreviation	Full name
CCMTS	Coastal container multimodal transportation system
CO_2	Carbon dioxide
STNP	Shipping network design—toll policy joint optimization problem of CCMTSs
ASA	Active set algorithm
NDP	Network design problem
TP	Toll problem
DNDP	Discrete network design problem
CNDP	Continuous network design problem
MILP	Mixed-integer linear programming
GA	Genetic algorithm
MPEC	Mathematical programming with equilibrium constraints
UE	User equilibrium
BPR	Bureau of Public Road
O-D	Origin - destination
WSN	Water subnetwork
LSN	Land subnetwork
STOM	Coastal container multimodal transportation system sustainable policy optimization model
CLF	Capacity limit factor
KKT	Karush-Kuhn-Tucker

The notations frequently used in this paper are listed as follows.

Sets	
$\mathscr{N}_s$	Set of ports
$\mathscr{N}_l$	Set of cites
$\mathscr{N}$	Set of all transport nodes, where $\mathscr{N} = \mathscr{N}_l \cup \mathscr{N}_s$
$\mathscr{A}_l$	Set of highway links
$\mathscr{A}_s$	Set of waterway links
$\mathscr{A}_m$	Set of connecting links
$\mathscr{A}$	Set of all links, where $\mathscr{A} = \mathscr{A}_l \cup \mathscr{A}_s \cup \mathscr{A}_m$
$\mathscr{W}$	Set of O-D (origin - destination) pairs
$\mathscr{P}_w$	Set of paths between O-D pair $w \in \mathscr{W}$
$A(p)$	Set of links along path $p \in \mathscr{P}_w$
Parameters	
a or (i,j)	Link $a = (i,j) \in \mathscr{A}$
B	Government budget ceiling
w	O-D pair $w \in \mathscr{W}$
q_w	Travel demand between O-D pair $w \in \mathscr{W}$
p	Path $p \in \mathscr{P}_w$
σ_{ap}^w	The path-link incidence, which equals 1 if path $p \in \mathscr{P}_w$ consists of link $a \in \mathscr{A}$ and 0 otherwise
g_a	Traffic volume capacity of link $a \in \mathscr{A}_l \cup \mathscr{A}_s$
Decision Variables	
x_a	Cargo flow volume on link $a \in \mathscr{A}$
γ_a	Government charges for unit goods on link $a \in \mathscr{A}_l \cup \mathscr{A}_s$
$y_{(a,k)}$	A $(0-1)$ variable, which equals 1 if the government increases the capacity of link a by k times the initial capacity
f_p^w	Cargo flow volume on path $p \in \mathscr{P}_w$, where $w \in \mathscr{W}$

3.4. Formulation of network and link cost functions

3.4.1. Network formulation

The CCMTS contains two transportation networks: a water subnetwork (WSN) and a land subnetwork (LSN). As shown in Fig. 2, cargos from City 3 to City 4 can be transported either by truck transport (i.e., Node 3 − Node 4) in only the LSN or by land-water transport in the two subnetworks (i.e., Node 3 − Node 1 − Node 2 − Node 4). We let $\mathscr{G}^s := (\mathscr{N}_s, \mathscr{A}_s)$ represent the WSN, where $\mathscr{N}_s$ denotes the set of ports, and $\mathscr{A}_s$ represents the set of waterway links. Similarly, $\mathscr{G}^l$ denotes the LSN, where $\mathscr{G}^l := (\mathscr{N}_l, \mathscr{A}_l)$. Here, $\mathscr{N}_l$ denotes the set of cities, and $\mathscr{A}_l$ represents the set of highway links. Note that a coastal city has both a city node and a port node (e.g., Rotterdam includes both the Rotterdam City and Port of Rotterdam). To address the above situation, we introduce both nodes for each coastal city. At the same time, these two nodes are connected using *connection links*. We use $\mathscr{A}_m$ to represent the set consisting of all connection links in $\mathscr{G} := \mathscr{G}^l \cup \mathscr{G}^s$. Let $\mathscr{W}$ denote the set of O-D pairs, the elements of which are denoted as w and let $\mathscr{A} :=$

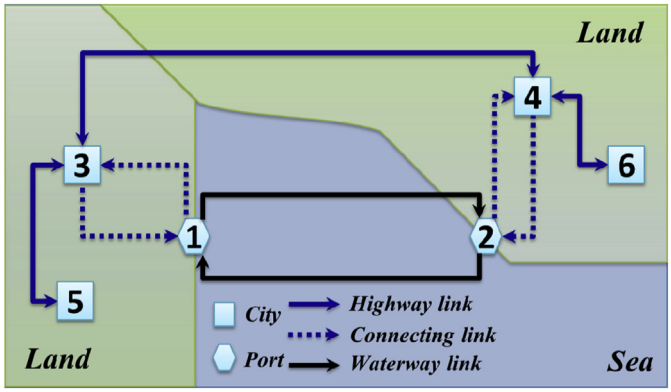


Fig. 2. CCMTS (Source: Chen et al. (2019)).

$\mathscr{A}_l \cup \mathscr{A}_s \cup \mathscr{A}_m, \mathscr{N} := \mathscr{N}_l \cup \mathscr{N}_s$. Note that for the waterway link $a \in \mathscr{A}_s$, due to the limited number of ships, there is an upper limit on transportation capacity.

3.4.2. Link cost (freight rate) functions

The link cost functions of different types of links are defined as follows. The link cost function of highway link a is shown in Constraint (1), where $\alpha_1, \alpha_2, \alpha_3, \alpha_4^a > 0$ are all parameters (see Meng and Wang (2011) for the same setting). Here, function $p_a(\cdot)$ represents the fright rate of unit cargo when passing through link a; d_a represents the distance between the head node and tail node of link a; x_a represents the cargo volume on link a; and s_a represents the vehicle average speed travelling on link a.

$$p_a(x_a) = \alpha_1 d_a + \frac{\alpha_2 d_a}{s_a}\left(\frac{x_a}{\alpha_4^a}\right)^{\alpha_3} \quad \forall a \in \mathscr{A}_l \tag{1}$$

The function $p_a(\cdot)$ of link $a \in \mathscr{A}_m$ is as shown in Constraint (2). Here, c_a^m represents the cost of loading (or unloading) the unit cargos to be paid at the port connected to link a. Therefore, the connection links do not exist in reality; they are used only to indicate the cargo loading (or unloading) activities and cost.

$$p_a = c_a^m \quad \forall a \in \mathscr{A}_m \tag{2}$$

In regard to waterway link a, function $p_a(x_a, y_{(a,k)})$ is as shown in Constraint (3). Here, $\beta_1, \beta_2, \beta_3^a > 0$ are both parameters, and c_a^s represents the basic freight rate of link $a \in \mathscr{A}_s$. Constraint (3) reflects the increase in freight for waterway links as cargo flows increase. It has the same form as the Bureau of Public Road (BPR) function.

$$p_a\left(x_a, y_{(a,k)}\right) = c_a^s\left[1 + \beta_1\left(\frac{x_a}{\beta_3^a}\right)^{\beta_2}\right] \quad \forall a \in \mathscr{A}_s \tag{3}$$

4. Model establishment

In this section, a mathematical model (also known as the 'coastal transport system sustainable policy optimization model', or STOM) is established (see Fig. 3).

STOM has an objective function of the lowest CO_2 emissions in the CCMTS. The STOM determines the cargo flow of each link, the government's shipping network design scheme (i.e., the waterway link capacity expansion scheme) and the government's optimal toll policy simultaneously. The constraints of STOM include government investment ceiling and the commitment to the shippers' *approximately Pareto-improving*.

4.1. Network equilibrium and approximately Pareto-improving

We first define the traffic network equilibrium (i.e., UE state) and approximately Pareto-improving principle of the CCMTS transportation network. Then, we give the mathematical expression of STOM.

Definition 1. At the equilibrium (i.e., UE) state, the freight rates of the utilized paths between an O-D pair are the same but less than or equal to that of any unutilized usable path between the same O-D pair (Sheffi, 1984).

Based on the classic Beckmann model (Sheffi, 1984), we can give the mathematical expression of the UE state as follows. Note that $\mathbf{Y} = (y_{(a,k)})$ is considered known in the model [UE].

$$[\text{UE}] \quad \min_{(\boldsymbol{X},\boldsymbol{f})} z_1 = \sum_{a \in \mathscr{A}} \int_0^{x_a} p_a(\omega) d\omega \tag{4}$$

$$s.t.\ x_a = \sum_{p \in \mathscr{P}_w} \sum_{w \in \mathscr{W}} f_p^w \sigma_{ap}^w \quad \forall a \in \mathscr{A} \tag{5}$$

$$x_a \le CA_a + y_{(a,1)} \cdot CA_a + 2y_{(a,2)} \cdot CA_a \quad \forall a \in \mathscr{A}_s \tag{6}$$

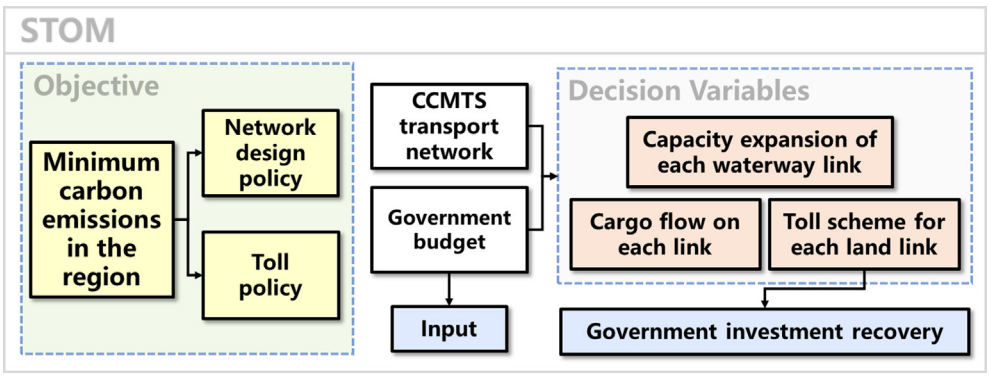


Fig. 3. The structure of STOM.

$$\sum_{p\in\mathscr{P}_w} f_p^w = q_w \quad \forall w\in\mathscr{W} \tag{7}$$

$$f_p^w \geq 0 \quad \forall p\in\mathscr{P}_w, w\in\mathscr{W} \tag{8}$$

$$y_{(a,k)}\in\{0,1\} \quad \forall a\in\mathscr{A}_s, k=1,2 \tag{9}$$

where q_w represents the transportation demand (O-D) between $w\in\mathscr{W}$; $\mathscr{P}_w$ denotes the path set between $w\in\mathscr{W}$, where $p\in\mathscr{P}_w$; f_p^w represents the cargo flow volume on path $p\in\mathscr{P}_w$; σ_{ap}^w is the path-link indicator variable, which equals 1 if path $p\in\mathscr{P}_w$ consists of link $a\in\mathscr{A}$ and 0 otherwise; $y_{(a,1)}$ and $y_{(a,2)}$ are two (0 −1) variables to indicate the degree of government expansion of link $a\in\mathscr{A}_s$.

In the above, Constraint (4) is the classic Beckmann model. Constraint (5) gives the relationship between x_a and f_p^w. Constraint (6) requires that the flow of each waterway link does not exceed its capacity. Recall assumption (iv) in Section 3.1; we assume that the government can expand the transportation capacity of link $a\in\mathscr{A}_s$ up to 4 times. For example, when $y_{(a,1)}=1$ and $y_{(a,2)}=0$, the capacity of $a\in\mathscr{A}_s$ will become $2CA_a$, which will be doubled. Similarly, when $y_{(a,1)}=0$ and $y_{(a,2)}=1$, the capacity of $a\in\mathscr{A}_s$ will be three times the original, that is, $3CA_a$. Note that the model proposed here is also applicable to the case of adding more link capacity. In general, we can use $\sum_{k=1}^{K}2^{k-1}y_{(a,k)}$, which varies between 0 and (2^K-1), where $y_{(a,k)}\in\{0,1\}$ for $k=1,2,\ldots,K$. Constraints (7) and (8) are basic constraints on network flows.

Proposition 1. *The above UE state exists and is unique.*

Proof. We first give the optimal conditions for the equilibrium state defined by Definition 1. According to Sheffi (1984), we form the Lagrangian of model [UE] as $\mathscr{L}(\mathbf{f},\mathbf{o},\mathbf{u}) = p[\mathbf{x}(\mathbf{f})]+\sum_{w\in\mathscr{W}}u_w\left(q_w-\sum_{p\in\mathscr{P}_w}f_p^w\right)+\sum_{a\in\mathscr{A}}o_a\left[CA_a\left(1+y_{(a,1)}+2y_{(a,2)}\right)-x_a\right]$. The first order conditions include $f_p^w\frac{\partial\mathscr{L}(\mathbf{f},\mathbf{o},\mathbf{u})}{\partial f_p^w}=0$, $\frac{\partial\mathscr{L}(\mathbf{f},\mathbf{o},\mathbf{u})}{\partial f_p^w}\geq 0$, $\frac{\partial\mathscr{L}(\mathbf{f},\mathbf{o},\mathbf{u})}{\partial u_w}=0$, $o_a\frac{\partial\mathscr{L}(\mathbf{f},\mathbf{o},\mathbf{u})}{\partial o_a}=0$ and $\frac{\partial\mathscr{L}(\mathbf{f},\mathbf{o},\mathbf{u})}{\partial o_a}\geq 0$. We then have the following optimal conditions.

(1)–(3), (5)–(9)

$$\left[CA_a\left(1+y_{(a,1)}+2y_{(a,2)}\right)-x_a\right]o_a=0 \quad \forall a\in\mathscr{A}_s \tag{10}$$

$$o_a\geq 0 \quad \forall a\in\mathscr{A}_s \tag{11}$$

$$f_p^w\left[\sum_{a\in A(p)\cap\mathscr{A}_m}p_a+\sum_{a\in A(p)\cap\mathscr{A}_l}p_a(x_a)+\sum_{a\in A(p)\cap\mathscr{A}_s,k=1,2}p_a\left(x_a,y_{(a,k)}\right)+\sum_{a\in\mathscr{A}_s}o_a\sigma_{ap}^w-u_w\right]=0 \quad \forall p\in\mathscr{P}_w, w\in\mathscr{W} \tag{12}$$

$$\sum_{a\in A(p)\cap\mathscr{A}_m}p_a+\sum_{a\in A(p)\cap\mathscr{A}_l}p_a(x_a)+\sum_{a\in A(p)\cap\mathscr{A}_s,k=1,2}p_a\left(x_a,y_{(a,k)}\right)+\sum_{a\in\mathscr{A}_s}o_a\sigma_{ap}^w-u_w\geq 0 \quad \forall p\in\mathscr{P}_w, w\in\mathscr{W} \tag{13}$$

where o_a is the Lagrangian multiplier (also known as 'Karush-Kuhn-Tucker multipliers' or 'KKT multipliers') associated with Constraint (6), which is called a capacity limit factor (CLF), reflecting the impact of a full load on $a\in\mathscr{A}_s$, and u_w is the KKT multiplier associated with Constraint (7), representing the equilibrium freight rate for O-D pair $w\in\mathscr{W}$.

In the above, Constraints (10) and (11) represent the complementary relationship between the full load and the CLF o_a of link $a\in\mathscr{A}_s$. In particular, if $CA_a(1+y_{(a,1)}+2y_{(a,2)})-x_a=0$, then $o_a\geq 0$. If $CA_a(1+y_{(a,1)}+2y_{(a,2)})-x_a>0$, then CLF o_a must equal 0 because there are no restrictions on capacity. Constraints (12) and (13) denote the complementary relationship between the flow of a path and its freight rate. When the cargo volume on path $p\in\mathscr{P}_w, w\in\mathscr{W}$ is greater than 0, say $f_p^w>0$, there must be $u_w=\sum_{a\in A(p)\cap\mathscr{A}_m}p_a+\sum_{a\in A(p)\cap\mathscr{A}_l}p_a(x_a)+\sum_{a\in A(p)\cap\mathscr{A}_s,k=1,2}p_a\left(x_a,y_{(a,k)}\right)+\sum_{a\in\mathscr{A}_s}o_a\sigma_{ap}^w$. If $f_p^w=0$, then there will be no restrictions on transportation cost for this path.

Obviously, the above optimal conditions are equivalent to the statement in Definition 1. When we further consider the cost of cargo loading/unloading as part of the cost of water transportation, then the *Hessian* of the link cost functions is positive definite. Thus, link flows are unique. Q. E. D.

Definition 2. (Approximately Pareto-improving). A government toll scheme $\gamma=(\gamma_a)$ is *approximately Pareto-improving* if and only if γ induces a cargo flow distribution that reduces the CO_2 emissions of the CCMTS, makes shippers slightly worse off by no more than $(1+\delta)$ times the state where no policy is implemented and whereby the average transportation cost does not exceed more than $(1+\mu)$ times the state where no policy is implemented.

We define the *approximate Pareto-improving* conditions in CCMTS as follows.

$$u_w\leq(1+\delta)u_w^{UE} \quad \forall w\in\mathscr{W} \tag{14}$$

$$\sum_{w\in\mathscr{W}}u_w/N_{O-D}\leq(1+\mu)\sum_{w\in\mathscr{W}}u_w^{UE}/N_{O-D} \tag{15}$$

where δ and μ are the growth rate coefficients, N_{O-D} is the number of O-D pairs in the CCMTS network, and u_w^{UE} is the equilibrium freight rate of O-D pair $w\in\mathscr{W}$ when no policies are implemented.

Proposition 2. *Considering both the toll scheme* $\gamma=(\gamma_a)$ *and network design policy* $\mathbf{Y}=(y_{(a,k)})$, *the optimal solution to the problem, say* $(\mathbf{X},\mathbf{Y},\mathbf{f},\mathbf{o},\gamma)^{\mathrm{T}}$, *exists.*

Proof. When the toll of all highway links is set to 0—i.e., $\gamma=(\gamma_a)=\mathbf{0}$—and the network design is not carried out (i.e., $\mathbf{Y}=\mathbf{0}$), it is also a feasible solution to the problem. Thus, the optimal solution to the problem exists because the feasible region is not empty. Q. E. D.

4.2. CCMTS sustainable policy optimization model

Based on the above discussion, we can propose a coastal transport system sustainable policy optimization model (i.e., model [STOM]) as follows.

$$[\text{STOM}] \quad \min_{(\mathbf{X},\mathbf{Y},\mathbf{f},\mathbf{o},\gamma)} z_2 = \sum_{a\in\mathscr{A}_l} e_l d_a x_a + \sum_{a\in\mathscr{A}_s} e_s d_a x_a \tag{16}$$

s.t. (1)–(3), (5)–(11), (14)–(15),

$$f_p^w \left[\sum_{a\in A(p)\cap\mathscr{A}_m} p_a + \sum_{a\in A(p)\cap\mathscr{A}_l} (p_a(x_a)+\gamma_a) + \sum_{a\in A(p)\cap\mathscr{A}_s,k=1,2} p_a\left(x_a, y_{(a,k)}\right) + \sum_{a\in\mathscr{A}_s} o_a \sigma_{ap}^w - u_w \right] = 0 \quad \forall p\in\mathscr{P}_w, w\in\mathscr{W} \tag{17}$$

$$\sum_{a\in A(p)\cap\mathscr{A}_m} p_a + \sum_{a\in A(p)\cap\mathscr{A}_l} (p_a(x_a)+\gamma_a) + \sum_{a\in A(p)\cap\mathscr{A}_s,k=1,2} p_a\left(x_a, y_{(a,k)}\right) + \sum_{a\in\mathscr{A}_s} o_a \sigma_{ap}^w - u_w \geq 0 \quad \forall p\in\mathscr{P}_w, w\in\mathscr{W} \tag{18}$$

$$\sum_{a\in\mathscr{A}_s} \varphi_a\left(y_{(a,1)} + 2y_{(a,2)}\right) \leq B \tag{19}$$

$$0 \leq \gamma_a \leq \gamma_{\max} \quad \forall a\in\mathscr{A}_l\cup\mathscr{A}_s \tag{20}$$

where $(\mathbf{X},\mathbf{Y},\mathbf{f},\mathbf{o}) \geq 0$; e_l and e_s are the CO_2 emission factors of the highway and waterway, respectively; B represents the budget ceiling of the government; φ_a represents the investment needed to add one ship (double the current capacity) of waterway link a; γ_a represents the government's toll for the unit cargos on link a; $\gamma_{\max}$ denotes the government's upper limit on the toll for unit cargo.

In the above, Equation (16) is the objective function that requires the government to minimize the CO_2 emissions in the CCMTS. Constraints (17) and (18) are improved from Constraints (12) and (13). They increase the decision variable γ_a to reflect the toll scheme set by the government. Constraint (19) requires that the government's shipping network design plan not exceed the established budget ceiling. When $y_{(a,1)} = 1$, that is, when the capacity of link a needs to be doubled, the left side of Constraint (19) will increase by φ_a. Similarly, when $y_{(a,2)} = 1$, the left side of Constraint (19) will increase by $2\varphi_a$, which is the cost of renting two more ships. Constraint (20) requires the government to toll no more than the upper limit (i.e., $\gamma_{\max}$) for each link.

4.3. Model for solving the UE state of the CCMTS transportation network

Recall that Constraints (14) and (15) require the equilibrium transportation cost between each O-D pair when the policy is not implemented as a parameter; therefore, solving the equilibrium state of CCMTS becomes necessary. To improve the efficiency of the solution, we further convert the problem [UE] in Section 4.1 to problem [UE-T] via the gap function proposed by Wu et al. (2011).

$$[\text{UE}-\text{T}] \quad \min_{(\mathbf{X},\mathbf{f},\boldsymbol{\kappa},\boldsymbol{\tau},\boldsymbol{\pi})} z_3 = \sum_{a\in\mathscr{A}} x_a p_a(x_a) - \sum_{a\in\mathscr{A}_s} CA_a\tau_a - \sum_{w\in\mathscr{W}} \pi_w q_w \tag{21}$$

$$s.t.\ x_a = \sum_{w\in\mathscr{W}} \sum_{p\in\mathscr{P}_w} f_p^w \sigma_{ap}^w \quad \forall a\in\mathscr{A} \tag{22}$$

$$x_a \leq CA_a \quad \forall a\in\mathscr{A}_s \tag{23}$$

$$\sum_{p\in\mathscr{P}_w} f_p^w = q_w \quad \forall w\in\mathscr{W} \tag{24}$$

$$\tau_a + \kappa_a \leq p_a\left(x_a, y_{(a,k)}\right) \quad \forall a\in\mathscr{A}_s, k=1,2 \tag{25}$$

$$\kappa_a \leq p_a(x_a) \quad \forall a\in\mathscr{A}\backslash\mathscr{A}_s \tag{26}$$

$$\sum_{w'\in\mathscr{W}} \pi_w \left(\sum_{p'\in\mathscr{P}_{w'}} \phi_{wp}^{w'p'} \right) - \sum_{a\in\mathscr{A}} \kappa_a \sigma_{ap}^w \leq 0 \quad \forall p\in\mathscr{P}_w, w\in\mathscr{W} \tag{27}$$

$$\tau_a \leq 0 \quad \forall a\in\mathscr{A}_s \tag{28}$$

where $(\mathbf{X},\mathbf{f}) \geq 0$, $\boldsymbol{\kappa} = (\kappa_a)$, $\boldsymbol{\tau} = (\tau_a)$ and $\boldsymbol{\pi} = (\pi_w)$ are the KKT multipliers associated with Constraints (22) to (24), and $\phi_{wp}^{w'p'}$ is the path-path $(0-1)$ indicator variable, which equals 1 if $p = p'$ $(p\in\mathscr{P}_w,\ p'\in\mathscr{P}_{w'},\ w,w'\in\mathscr{W})$ and 0 otherwise.

According to Aghassi et al. (2006), the optimal solution $(\mathbf{X}^*,\mathbf{f}^*,\boldsymbol{\kappa}^*,\boldsymbol{\tau}^*,\boldsymbol{\pi}^*)^{\mathrm{T}}$ of problem [UE-T] solves a cargo flow distribution that satisfies Definition 1 if and only if z_3 equals 0. Note that problem [UE-T] is only a regular nonlinear programming model and can be easily solved by commercial solvers (e.g., CONOPT (Drud, 1995) and CPLEX (CPLEX, 2008)).

5. Solution algorithm

As previously discussed, STOM has *equilibrium conditions* (see Constraints (17)-(18)). Therefore, solving global optimal solutions for this type of model will become especially difficult (see Luo et al. (1996) and Chen et al. (2016) for two examples). In this section, we propose an ASA developed from Lawphongpanich and Yin (2010) to solve the local optimal solution of [STOM].

The design idea of the ASA that we proposed is first to solve problem [R-STOM] (a problem with active sets) by relaxing model [STOM] and obtain the KKT multipliers associated with the constraints related to these sets. The active sets in problem [R-STOM] can be updated based on these multipliers. Then, we resolve model [R-STOM] based on updated sets. Finally, an local optimal WSN design scheme and a toll policy implementation scheme can be obtained simultaneously when iteration stops.

To apply ASA, we first define the following active sets and problem [R-STOM]. Let $\Gamma_W := \{(w,p) : f_p^w = 0, w\in\mathscr{W}, p\in\mathscr{P}_w\}$, $\Gamma_{\tilde{W}} := \{(w,p) : h_p^w = 0, w\in\mathscr{W}, p\in\mathscr{P}_w\}$, $\Gamma_A := \{a : o_a = 0, a\in\mathscr{A}_s\}$, and $\Gamma_{\tilde{A}} := \{a : CA_a(1 + y_{(a,1)} + 2y_{(a,2)}) - x_a = 0, a\in\mathscr{A}_s\}$, where $\Gamma_W\cap\Gamma_{\tilde{W}} = \varnothing$, $\Gamma_W\cup\Gamma_{\tilde{W}} = \mathscr{A}\backslash\mathscr{A}_m$, $\Gamma_A\cap\Gamma_{\tilde{A}} = \varnothing$ and $\Gamma_A\cup\Gamma_{\tilde{A}} = \mathscr{A}_s$. We then establish problem [R-STOM] as below.

$$[\text{R}-\text{STOM}] \quad \min_{(\mathbf{X},\mathbf{Y},\mathbf{f},\mathbf{o},\gamma)} z_4 = \sum_{a\in\mathscr{A}_l} e_l d_a x_a + \sum_{a\in\mathscr{A}_s} e_s d_a x_a \tag{29}$$

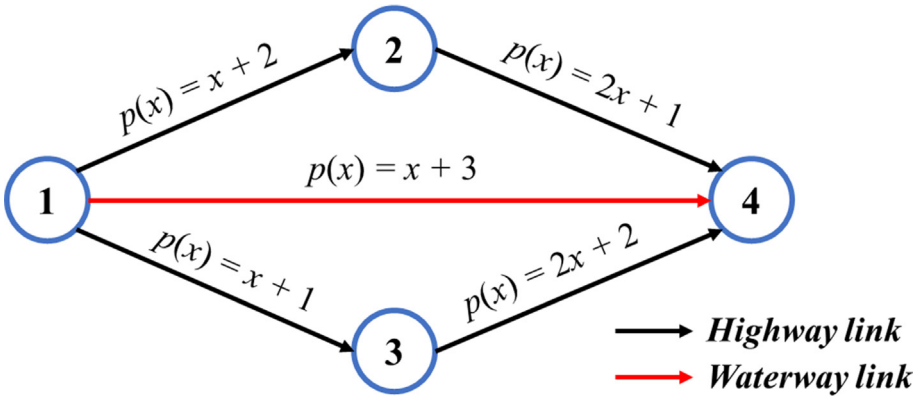


Fig. 4. A small-scale transportation network.

Table 1
Cargo flow on different paths and total CO_2 emissions in different scenarios.

Scenario	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10
Toll on link 1-2	0.00	3.00	3.00	3.00	5.00	5.00	5.00	10.00	10.00	10.00
Toll on link 1-3	0.00	3.00	5.00	10.00	3.00	5.00	10.00	3.00	5.00	10.00
Cargo flow on Path 1	2.00	1.40	1.53	1.75	0.87	1.00	1.25	0.00	0.00	0.00
Cargo flow on Path 2	6.00	7.20	7.60	8.25	7.60	8.00	8.75	8.25	8.75	10.00
Cargo flow on Path 3	2.00	1.40	0.87	0.00	1.53	1.00	0.00	1.75	1.25	0.00
Total CO_2 emission	14.00	12.80	12.40	10.00	12.40	12.00	11.25	11.75	11.25	10.00

s.t. (1)–(3), (5)–(9), (14)–(15), (19)–(20),

$$h_p^w = \sum_{a\in A(p)\cap\mathscr{A}_m} p_a + \sum_{a\in A(p)\cap\mathscr{A}_l} (p_a(x_a) + \gamma_a) + \sum_{a\in A(p)\cap\mathscr{A}_s,k=1,2} p_a\left(x_a, y_{(a,k)}\right) + \sum_{a\in\mathscr{A}_s} o_a\sigma_{ap}^w - u_w \quad \forall p\in\mathscr{P}_w, w\in\mathscr{W} \tag{30}$$

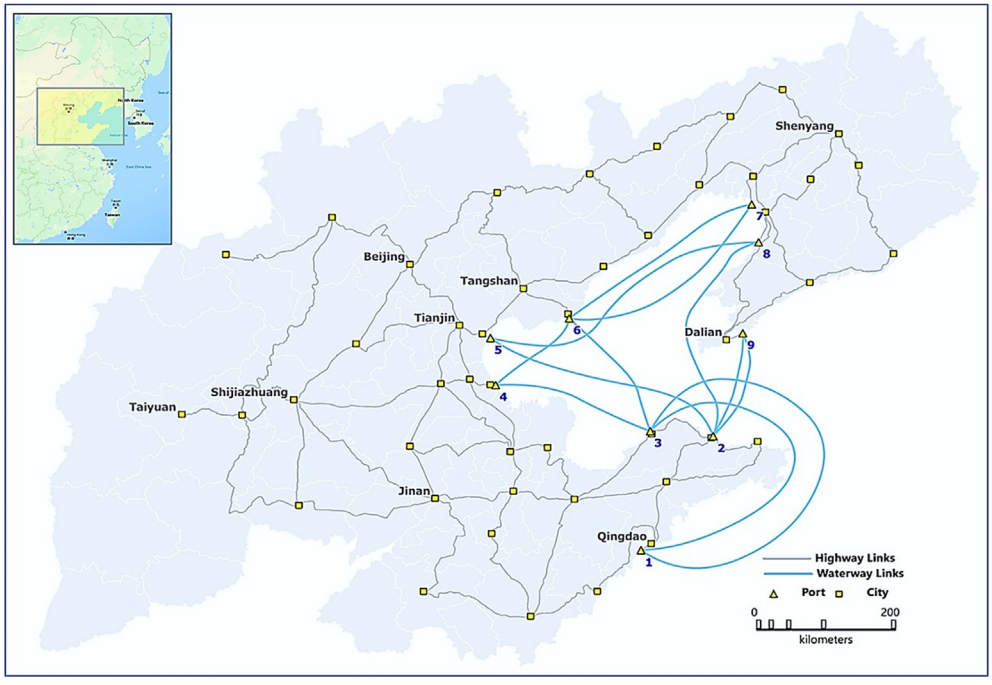


Fig. 5. Transportation network used in the experiment (Source: Chen et al. (2019)).

Table 2
Linear routes used in the experiments.

No.	Initial capacity (TEU)	Routes
1	500	(7, 6), (6, 3), (3, 1), (1, 3), (3, 4), (4, 6), (6, 7)
2	200	(9, 2), (2, 9)
3	400	(8, 2), (2, 5), (5, 8)

$$f_p^w = 0 \quad \forall (w,p) \in \Gamma_W \tag{31}$$

$$h_p^w \geq 0 \quad \forall (w,p) \in \Gamma_W \tag{32}$$

$$f_p^w \geq 0 \quad \forall (w,p) \in \Gamma_{\tilde{W}} \tag{33}$$

$$h_p^w = 0 \quad \forall (w,p) \in \Gamma_{\tilde{W}} \tag{34}$$

$$o_a = 0 \quad \forall a \in \Gamma_A \tag{35}$$

$$CA_a\left(1 + y_{(a,1)} + 2y_{(a,2)}\right) - x_a \geq 0 \quad \forall a \in \Gamma_A \tag{36}$$

$$o_a \geq 0 \quad \forall a \in \Gamma_{\tilde{A}} \tag{37}$$

$$CA_a\left(1 + y_{(a,1)} + 2y_{(a,2)}\right) - x_a = 0 \quad \forall a \in \Gamma_{\tilde{A}} \tag{38}$$

Based on the above settings, we give the specific algorithm process for solving problem [R-STOM].

Step 1: Set $\vartheta = 1$, $\gamma = (\gamma_a) = \mathbf{0}$, $Y = (y_{(a,k)}) = 0$. Let $\Gamma_W^\vartheta := \left\{(w,p) : f_p^w = 0, w \in \mathscr{W}, p \in \mathscr{P}_w\right\}$, $\Gamma_{\tilde{W}}^\vartheta := \left\{(w,p) : h_p^w = 0, w \in \mathscr{W}, p \in \mathscr{P}_w\right\}$, $\Gamma_A^\vartheta := \{a : o_a = 0, a \in \mathscr{A}_s\}$ and $\Gamma_{\tilde{A}}^\vartheta := \left\{a : CA_a\left(1 + y_{(a,1)} + 2y_{(a,2)}\right) - x_a = 0, a \in \mathscr{A}_s\right\}$ according to the solution of problem [UE-T] and go to **Step 2**.
Step 2: Let $(\mathbf{X}^\vartheta, \mathbf{Y}^\vartheta, \mathbf{f}^\vartheta, \mathbf{o}^\vartheta, \gamma^\vartheta)^{\mathrm{T}}$ solve the following model and go to **Step 3**.

$$\min_{(\mathbf{X},\mathbf{Y},\mathbf{f},\mathbf{o},\gamma)} z_5 = \sum_{a \in \mathscr{A}_l} e_l d_a x_a + \sum_{a \in \mathscr{A}_s} e_s d_a x_a \tag{39}$$

s.t. (1)–(3), (5)–(9), (14)–(15), (19)–(20),

$$f_p^w = 0 \quad \forall (w,p) \in \Gamma_W^\vartheta \tag{40}$$

$$h_p^w \geq 0 \quad \forall (w,p) \in \Gamma_W^\vartheta \tag{41}$$

$$f_p^w \geq 0 \quad \forall (w,p) \in \Gamma_{\tilde{W}}^\vartheta \tag{42}$$

$$h_p^w = 0 \quad \forall (w,p) \in \Gamma_{\tilde{W}}^\vartheta \tag{43}$$

$$o_a = 0 \quad \forall a \in \Gamma_A^\vartheta \tag{44}$$

$$CA_a\left(1 + y_{(a,1)} + 2y_{(a,2)}\right) - x_a \geq 0 \quad \forall a \in \Gamma_A^\vartheta \tag{45}$$

$$o_a \geq 0 \quad \forall a \in \Gamma_{\tilde{A}}^\vartheta \tag{46}$$

$$CA_a\left(1 + y_{(a,1)} + 2y_{(a,2)}\right) - x_a = 0 \quad \forall a \in \Gamma_{\tilde{A}}^\vartheta \tag{47}$$

Step 3: Let $\Lambda_W^\vartheta := \{(w,p) \in \Gamma_W^\vartheta \cap \Gamma_{\tilde{W}}^\vartheta : \eta_\vartheta^{w,p} < 0\}$ and $\Lambda_A^\vartheta := \{a \in \Gamma_A^\vartheta \cap \Gamma_{\tilde{A}}^\vartheta : \tilde{\eta}_\vartheta^a < 0\}$, where $\eta_\vartheta^{w,p}$ and $\tilde{\eta}_\vartheta^a$ are the KKT multipliers associated with Constraints (40) and (47), repectively. If $\Lambda_W^\vartheta \cup \Lambda_A^\vartheta = \varnothing$, stop, and set $(\mathbf{X}^\vartheta, \mathbf{Y}^\vartheta, \mathbf{f}^\vartheta, \mathbf{o}^\vartheta, \gamma^\vartheta)^{\mathrm{T}}$ is the local optimal solution of problem [R-STOM]; otherwise, go to **Step 4**.
Step 4: Set $\Gamma_W^{\vartheta+1} = \Gamma_W^\vartheta - \Lambda_W^\vartheta$, $\Gamma_A^{\vartheta+1} = \Gamma_A^\vartheta - \Lambda_A^\vartheta$; $\Gamma_{\tilde{W}}^{\vartheta+1} = \{(w,p) : h_p^w = 0, w \in \mathscr{W}, p \in \mathscr{P}_w\}$, $\Gamma_{\tilde{A}}^{\vartheta+1} = \{a : CA_a(1 + y_{(a,1)} + 2y_{(a,2)}) - x_a = 0, a \in \mathscr{A}_s\}$ and go to **Step 5**.
Step 5: Let $\vartheta = \vartheta + 1$ and go to **Step 2**.

Note that at **Step 1**, the optimal solution of problem [R-STOM] always provides a feasible solution to problem [R-STOM] when $\gamma^1 = \mathbf{0}$ and $\mathbf{Y} = \mathbf{0}$. We choose sets $(\Gamma_W^1, \Gamma_{\tilde{W}}^1)$ and $(\Gamma_A^1, \Gamma_{\tilde{A}}^1)$ to ensure that $(\mathbf{X}^1, \mathbf{Y}^1, \mathbf{f}^1, \mathbf{o}^1, \gamma^1)$ is feasible. For analysis of algorithm efficiency, please refer to Zhang et al. (2009) and Chen et al. (2016).

6. Numerical experiments

In this section, we first use a small-scale example to prove the effectiveness of ASA. We then solve a real-scale problem and carry out a sensitivity analysis to verify the effectiveness of the model. The computer used in the experiments is produced by *Lenovo*, having an Intel Core i7-8550U processor (i.e., a 4.0 GHz CPU) and 12.0 GB of RAM.

6.1. Numerical experiment on a small-scale network

Recall that the model [STOM] proposed in this paper introduces transportation capacity constraints, which makes equilibrium analysis and problem solving more difficult. On the one hand, the capacity constraint adds the generalized Lagrangian multiplier o_a (i.e., CLF) to the equilibrium conditions, making it impossible to solve the problem using the piecewise linearization method proposed by Luathep et al. (2011). On the other hand, the enumeration of continuous variables quickly increases the scale of the problem. To illustrate the efficiency of the application of ASA in this paper, we use a small-scale experiment to explain further.

As shown in Fig. 4, a toy transportation network consisting of four cities is proposed. The red line indicates the waterway link, and

Table 3
Parameter settings.

Parameter	Value of parameter	Reference
α_1	1	Meng and Wang (2011)
v_c	23	
α_2	4	
α_3	2	
c_a^m	4	
c_a^s	0.5	
β_1	2	
β_2	2	
e_l	1.24	Chen et al. (2014)
e_s	0.32	
δ	1.05	Assumption (iii) in Section 3.1
μ	1.025	
ξ	0.0001	–
$\gamma_{\max}$	10000	

Table 4
Sensitivity experiment 1 calculation results.

No.	B ($\times 10^3$ RMB/week)	Total toll ($\times 10^3$ RMB/week)	Total government investment ($\times 10^3$ RMB/week)	Government net expenditure ($\times 10^3$ RMB/week)	Total emissions of CCMTS when policies are not implemented ($\times 10^7$ ton/week)	Total CO_2 emissions of CCMTS when policies are implemented ($\times 10^7$ ton/week)	Running time (s)	Emission reduction ratio (%)
(1)	(2)	(3)	(4)	(5) = (4)−(3)	(6)	(7)	(8)	(9) = [(6)−(7)]/(7)
B1	300	767.703	297.603	−470.100	1.163	1.126	48.30	3.286%
B2	450	581.663	413.848	−167.816		1.106	48.12	5.154%
B3	600	681.757	599.307	−82.450		1.095	64.72	6.210%
B4	750	720.477	663.955	56.522		1.092	50.52	6.502%
B5	900	715.665	847.133	131.469		1.090	48.17	6.697%
B6	1050	715.665	1019.483	303.818		1.090	48.35	6.697%

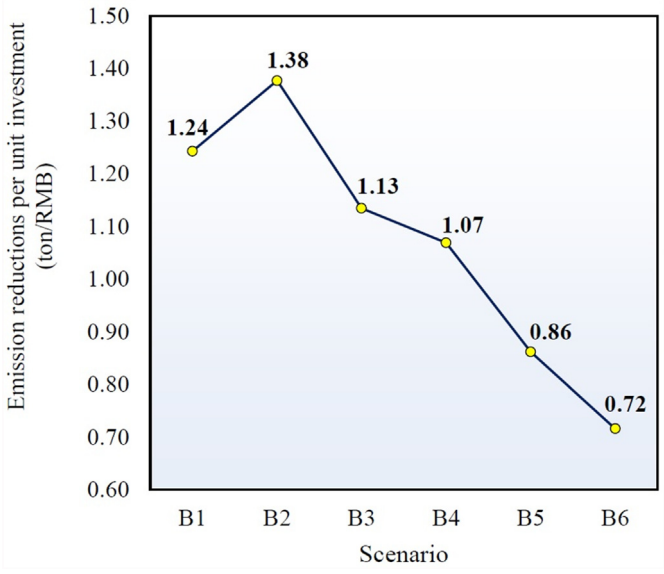


Fig. 6. Emission reductions by 1 RMB in scenarios B1 ~ B6.

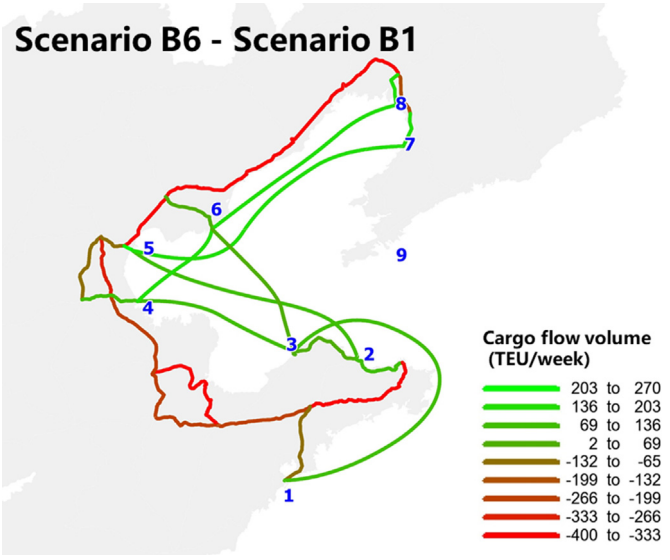


Fig. 7. The difference between the cargo flow distribution of scenarios B6 and B1.

Table 5
The government shipping network design plans for scenarios B1 to B6.

Scenario	2× capacity expansion of waterway link ($y_{(a,1)} = 1$, $y_{(a,2)} = 0$)	3× capacity expansion of waterway link ($y_{(a,1)} = 0$, $y_{(a,2)} = 1$)	4× capacity expansion of waterway link ($y_{(a,1)} = 1$, $y_{(a,2)} = 1$)
B1	(2, 9), (4, 6)	(7, 6)	
B2	(3, 4), (4, 6), (6, 7),	(7, 6)	
B3	(5, 8), (2, 9), (3, 4), (4, 6), (6, 7)	(7, 6)	
B4	(5, 8), (2, 9), (3, 4), (4, 6), (6, 7), (6, 3)	(7, 6)	
B5	(2, 5), (5, 8), (2, 9), (4, 6), (6, 7), (6, 3)	(2, 9), (7, 6)	
B6	(2, 5), (5, 8), (2, 9), (4, 6), (6, 7), (6, 3)		(3, 4), (7, 6)

the black line indicates the highway link. The cargo from City 1 to City 4 has three transportation routes, namely Path 1: 1-2-4, Path 2: 1–4 and Path 3: 1-3-4. Assume the CO_2 emissions per unit of cargo for the three paths are 2, 1 and 2, respectively. The relationship between the freight rate and cargo flow of each link is located above the link in Fig. 4. Suppose there are 10 units of cargos that need to be transported from City 1 to City 4. At the same time, the waterway link has a capacity limit of 10 units of cargo. Based on the above settings, we can calculate the cargo flow for each path in the equilibrium state as $(2, 6, 2)^T$ and the CO_2 emissions as 14. We record the above situation as scenario T1.

When considering the toll scheme, the problem becomes complicated. To solve the above problem, the enumeration method will be introduced. We assume that links 1–2 and 1–3 can charge 3, 5 and 10 per unit cargo, so there will be nine different scenarios (denoted as T2 to T10). Table 1 shows the cargo flow and CO_2 emissions in the equilibrium state in different scenarios. When the tolls for both links are set to 10, the carbon emission of the transportation system is the lowest at 10. Similarly, when we introduce constraints that require two links to charge between 0 and 10 and use ASA to solve the model, the optimal solution obtained is the same. This validates the performance of the ASA algorithm in solving small-scale problems. Note that even for such a small-scale case, it is very difficult to solve using the enumeration method

Table 6
Sensitivity experiment 2 calculation results.

No.	O-D demand	Total toll ($\times 10^3$ RMB/week)	Total government investment ($\times 10^3$ RMB/week)	Government net expenditure ($\times 10^3$ RMB/week)	Total emissions of CCMTS when policies are not implemented ($\times 10^7$ ton/week)	Total CO_2 emissions of CCMTS when policies are implemented ($\times 10^7$ ton/week)	Running time (s)	Emission reduction ratio (%)
(1)	(2)	(3)	(4)	(5) = (4)−(3)	(6)	(7)	(8)	(9) = [(6)−(7)]/(7)
OD1 (B1)	100%	767.703	297.603	−470.100	1.163	1.126	48.30	3.286%
OD2	150%	620.469	297.120	−323.349	1.886	1.831	44.25	3.004%
OD3	200%	2488.374	297.120	−2191.254	2.629	2.539	44.96	3.545%
OD4	250%	1567.872	265.244	−1302.628	3.371	3.275	43.57	2.931%
OD5	300%	1716.513	265.244	−1451.268	4.118	3.967	45.10	3.806%

Table 7
The government shipping network design plans for scenarios OD1 to OD5.

Scenario	2× capacity expansion of waterway link ($y_{(a,1)} = 1, y_{(a,2)} = 0$)	3× capacity expansion of waterway link ($y_{(a,1)} = 0, y_{(a,2)} = 1$)
OD1 (B1)	(2, 9), (4, 6)	(7, 6)
OD2	(5, 8), (8, 2), (9, 2)	
OD3	(5, 8), (8, 2), (9, 2)	
OD4	(8, 2), (2, 9)	(9, 2)
OD5	(8, 2), (2, 9)	(9, 2)

because the accuracy of the value of continuous variables will affect the quality of the optimal solution. Therefore, in the following section, we will use only ASA to address the problem in actual scale.

6.2. Data and parameter settings

The CCMTS transportation network used is located in Northeast China (see Fig. 5 for details).

In the Bohai Rim, 46 cities and 136 highway links are densely distributed, forming the nodes and arcs of the CCMTS's LSN. In regard to the WSN, there are 12 waterway links, and the liners provide transportation services among 9 ports. Liner routes can be found in Table 2 and one container ship of different sizes depart every week on each route. The port names and their corresponding numbers are Qingdao (No. 1), Yantai (No. 2), Longkou (No. 3), Huanghua (No. 4), Tianjin (No. 5), Tangshan (No. 6), Panjin (No. 7), Yingkou (No. 8) and Dalian (No. 9). Additionally, there are 18 connecting links in the CCMTS transport network that connect the port and coastal city.

The transportation demand (O-D) data come from the container transportation volume in the Bohai Rim region of China. The data are set based on a survey that we conducted. The unit of transportation demand is a 'twenty-feet equivalent unit' (TEU), and the time unit is a week. We assume that the equilibrium freight rate between any two cities cannot exceed 102.5% of the time when the toll policy is not implemented, and the average equilibrium freight rate cannot exceed 101.25% when the toll policy is not implemented. In addition, we refer to Meng and Wang (2011) and Chen et al. (2014) for the parameter settings (see Table 3 for details).

6.3. Sensitivity analysis

6.3.1. Sensitivity analysis of government investment

We set different levels of government budget ceilings and calculate the optimal WSN design scheme and toll policy scheme for CO_2 emission reductions. We calculate the government's optimal policy implementation plan under the conditions of 300,000 RMB/week to 1050,000 RMB/week with a step of 150,000 RMB. The results are shown in Table 4.

In Table 4, the 'total toll' is obtained by multiplying the toll of each link $a \in \mathscr{A}_l$ by the weekly cargo flow volume on each link and then summing them. The 'total government investment' is obtained as the actual expenditure of the government when investing in the expansion of the waterway link. The 'government net expenditure' is the difference between the total government investment and the total toll. The 'total emissions of CCMTS when policies are not implemented' is measured by the total emissions of CO_2 in the CCMTS before the government implemented any policy. The 'total CO_2 emissions of CCMTS when policies are implemented' indicates the CO_2 emissions of the CCMTS when the government implements the sustainable policy. The 'running time' refers to the program running time. The 'emission reduction ratio' refers to the percentage reduction of CO_2 emissions after the implementation of the sustainable policy.

The results in Table 4 are in line with our expectations. The running time of the program for the full-scale problem usually does not exceed 1 minute. This shows that the algorithm we proposed can efficiently handle a medium-to large-scale problem. From the results, as the government investment ceiling gradually increases, the total amount of government toll under different scenarios fluctuates approximately 700,000 RMB. At the same time, the total regional carbon emissions decline from 0.037×10^7 ton/week to 0.073×10^7 ton/week in scenario B6. Fig. 6 shows the carbon emissions that can be eliminated per unit investment in different scenarios. It shows well the principle of 'decreasing marginal returns' in economic theory. In the case of limited funds, the optimal government investment scale should be 450,000 RMB/week. Table 5 shows the government shipping network design plans for scenarios B1 ~ B6.

In scenario B1, due to the limited amount of government investment, the transport capacity of the 'Dalian-Yantai' and 'Tangshan-Huanghua' routes can both be expanded to 2 times the original capacity, and the capacity of the 'Panjin-Yingkou' route can be expanded to 3 times the original capacity. In regard to Scenario B6, the capacities of 8 waterway links can be expanded to varying degrees because the government budget is sufficient. It can be found in different scenarios that link (7, 6) should give priority to its expansion. The above results indicate that the bottleneck of reducing CO_2 emissions under the current O-D demand lies in the 'Panjin-Tangshan' route. Fig. 7 shows the difference in cargo flow

distribution in scenarios B6 and B1. The green to red gradient indicates that the cargo flow goes from rising to falling. As shown in Fig. 7, waterway links appear green, while highway links appear red. As the government investment level grew, the regional CO_2 emission reduction rate increased from 3.286% to 6.697% as more cargo could be transported by water.

6.3.2. Sensitivity analysis of transportation demand (O-D data)

In this subsection, a sensitivity analysis is conducted for container transportation demand (O-D data). The government policy implementation plans under 150%, 200%, 250% and 300% of scenario B1 are calculated. The results are shown in Table 6. Note that the meaning of each column in Table 6 is the same as in Table 4.

Table 6 also agrees with the actual situation. With the continuous rise of O-D demand in the Bohai Rim Region, the government's toll for highway links will gradually increase, from the 100,000 level in scenario OD1 to the million level in scenario OD5. Because the government's investment scale (parameter B) remains unchanged, the CO_2 emission reduction rate of the CCMTS transportation network is maintained at approximately 3%. Therefore, under the premise of limited funds, the flexible use of toll policy has become an important way for the government to control emissions. Table 7 shows the government shipping network design schemes for scenarios OD1 ~ OD5.

As shown in Table 7, the emphasis of the government's shipping network design plans is different in various scenarios; however, links (2, 9), (9, 2) and (8, 2) have always played an important role. Due to the limited capacity of links (2, 9) and (9, 2), increasing the capacity of these links will further guide the use of water transport. Although the initial capacity of link (8, 2) is high, it also becomes a bottleneck in the transportation network as the O-D data increase. The above phenomenon indicates that Yingkou Port has an important position in regional emission reduction. In reality, Yingkou Port undertakes a large number of China's Bohai Rim domestic trade container shipping every year. The above results further prove that the model and algorithm in this paper are effective and efficient.

6.4. Planning implications

From the above analysis, we can draw the following main planning implications.

First, the Dalian-Yantai transportation channel is the bottleneck in the Bohai Rim region. For the government, on the one hand, the construction of waterway transportation channels should be strengthened. The government can consider appropriate investment to increase the frequency of liner shipping and switch to large ships to improve the customer service level of the water carrier. The government can also reduce transportation costs through appropriate subsidy policies. At the same time, the government should advocate the establishment of the Dalian-Shandong land transportation corridor. The construction of the Bohai Strait Cross-sea corridor will transform the Bohai Rim from the original C-shaped transportation to I-shaped direct transportation. Reducing the transport distance will significantly reduce the carbon dioxide emissions of road transport. Actually, as of February 2019, the construction plan of the Bohai Strait Cross-sea corridor has appeared in various national and local documents.

Second, the land transportation between Tangshan and Panjin is the current key emission route. As a transportation gateway in Liaoning Province, Panjin undertakes a large number of land transportation transit tasks. Therefore, the Tangshan-Panjin land transport will bring a significant amount of carbon dioxide emissions to the region. The government should strengthen the waterway transportation capacity of the two cities. In practice, the government can formulate reasonable incentive policies to promote shippers to choose water transportation in the transportation of related cities and further reduce carbon emissions.

Third, the port of Yingkou plays an important role in the Bohai Rim region. To reduce regional carbon dioxide emissions, the government should support domestic container shipping in Yingkou Port. The government can arrange a moderate-scale investment and add a short-haul route from Yingkou to other important ports around the Bohai Sea. It can also provide the necessary subsidies for shippers of domestic trade container transportation to incentivize them to realize the transformation of transportation methods.

7. Conclusion

This paper analyzes three stakeholders' behaviors in the CCMTS. Unlike previous literature, we address the impact of decision-making among the three stakeholders and the efficiency of the solving algorithm. From the perspective of the government, a programming model is proposed (short as 'STOM') to jointly explore the government's optimal network design policy and toll scheme. An efficient algorithm for solving the local optimal solution of the abovementioned model is also proposed. The real-world container shipping data of Northeast China are used to verify the validity of the model. The results show the following phenomena.

(i) Under different budget ceilings, the government can reduce the CO_2 emissions in the CCMTS by 3.29%—6.70%. At the same time, the CO_2 reduction per unit investment decreased from 1.38 ton/RMB to 0.72 ton/RMB as the investment increased. This result illustrates the diminishing marginal returns in government investment. From the sensitivity analysis of various budget ceilings, we find that the optimal government investment scale should be maintained at 599.307×10^3 RMB/week.

(ii) Many studies of network design note that improving the traffic capacities of the key links enhances policy performance (see Chen et al. (2018) for an example). A similar phenomenon is also observed in our numerical experiments. The 'Dalian-Yantai' waterway link is the key route (bottleneck) for reducing CO_2 emissions as O-D demand increases. The important position of Yingkou Port in the Bohai Rim region can also be found. This result is consistent with the current policy orientation. To achieve low-carbon transportation, the Chinese government is demonstrating the feasibility of the 'Dalian-Yantai' subsea tunnel construction. In addition, Yingkou Port currently undertakes the transfer of a large number of domestic trade containers.

(iii) With a slight increase in freight rates between two cities (not more than 102.5% when the policy is not implemented), the government's toll policy can recover all investment while providing additional tax revenue for the government in small- and medium-sized investment and O-D demand level.

By and large, our findings can provide the necessary theoretical basis for the government. The model and algorithm we proposed are also applicable to other countries. After the local government knows the transportation demand data of the area, it can use the theory of this paper to design and evaluate a network design and toll policy. Finally, this paper can be further expanded. For example, we do not allow the government to invest in shipping routes that are not currently operational in the model. In addition, the O-D transportation demands in this paper are fixed. An in-depth discussion of the STNP considering the uncertainty O-D demand may also be an interesting research direction in the future.

CRediT authorship contribution statement

Zhongzhen Yang: Conceptualization, Methodology, Writing - original draft, Formal analysis, Writing - review & editing, Funding acquisition. **Xu Xin:** Methodology, Software, Formal analysis, Writing - original draft, Writing - review & editing. **Kang Chen:** Methodology, Software, Supervision, Funding acquisition, Writing - original draft, Writing - review & editing. **Ang Yang:** Investigation, Methodology, Data curation, Software, Writing - original draft, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors would like to express their gratitude for the support given by the National Natural Science Foundation of China [grant numbers 71431001 and 71402013], the Social Science Planning Foundation of Liaoning Province [grant number L19BGL005], the Natural Science Foundation of Liaoning Province [grant number 2020-HYLH-39] and the Fundamental Research Funds for the Central Universities [grant number 3132020234].

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